Practice Guide 03_01: Classic Anomaly Detectors on Network Connections

Aprendizagem Aplicada à Segurança

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Objective

Build the statistical core of an anomaly detector **yourself**: the z-score, the Mahalanobis distance, a threshold set on normal data only, and average precision; compare each with its scikit-learn twin. Then run the classic detectors of the lecture on KDD Cup 99 and measure why a classifier is not enough for attacks it has never seen.

- **Lecture:** `slides/slides_03_01.pdf`; **demos:** `notebooks/02-anomaly/` 00 to 03.
- **Notebook to complete:** `guide_03_01.ipynb` (this folder). Cells marked `TODO` are yours; `check` cells tell you whether you are right. The shared helpers come from `notebooks/02-anomaly/anomlib.py`.
- **Data:** Example 1 (generated) and `datasets/KDDCup99.csv.gz` (10% file, deduplicated to 145,585 connections).
- **Time:** about 2 hours (suggested: A 5 min, B 20, C 20, D 25, E 10, F 25, questions 15).
- **Environment:** `make venv` at the root of the repository; no GPU needed.

Background in one page

Setting. Train on normal rows only (*novelty detection*); a detector returns a score $s(\mathbf{x})$, higher = more anomalous, and alarms when $s(\mathbf{x}) > t$.

z-score and Mahalanobis. $z_j = (x_j - \mu_j)/\sigma_j$ and $s = \max_j |z_j|$ looks at one feature at a time. The Mahalanobis distance uses the covariance:

$$d_M(\mathbf{x})^2 = (\mathbf{x} - \mu)^\top C^{-1} (\mathbf{x} - \mu)$$

For Gaussian data $d_M^2 \sim \chi_d^2$. With many one-hot features S is singular: shrink it, $\hat{C} = (1-\lambda)S + \lambda\frac{\text{tr}(S)}{d}I$ (Ledoit-Wolf chooses λ from the data).

Threshold. t = the $(1-\alpha)$ quantile of the scores of **normal validation** rows: about α false alarms, no attack needed.

Base rate. With a share b of attacks, $P(\text{attack} \mid \text{alarm}) = \frac{\text{TPR}\, b}{\text{TPR}\, b + \text{FPR}\,(1-b)}$.

Average precision (AUPRC). Rank the rows by score; at the rank of each attack, compute the precision of the rows ranked above it (inclusive); average. A random score has AP equal to the attack share.

Isolation Forest. $s = 2^{-E[h]/c(n)}$, $c(n) = 2H(n-1) - 2(n-1)/n$, $H(i) \approx \ln i + 0.5772$; h = number of random cuts to isolate the point.

Tasks

Part	What you do	Check
A	Load Example 1 and KDD99, split (given)	
B	Implement `zmax` and `mahalanobis2`	lecture numbers, scikit-learn twin
C	Implement `threshold`, `average_precision`, `precision_at_base_rate`	`average_precision_score`
D	Implement `fit_shrunk`; compare with `LedoitWolf`; run the zoo (given)	table
E	Implement `c(n)` and `iforest_score`	$c(256) = 10.245$
F	Implement `train_without(category)`: a random forest that never saw one attack kind	table vs PCA

Tips. Use matrix operations (`np.einsum`) instead of loops over rows. Run the `check` cell after each part. If a check fails, print shapes and a few values before changing code.

Questions

1. **(B1)** Which of A and B does each score flag, and why? Sketch the contour $\max|z| = 2$ and the ellipse $d_M = 3$ over Example 1.
2. **(C1)** For TPR 0.95, how many of 100 alarms are real at a base rate of 1/10,000 and FPR 1%? And at FPR 0.1%? What does this mean for a security operations centre?
3. **(C2)** Why can the threshold be set without any attack? What is the risk if the validation rows contain attacks?

4. **(D1)** Which category is hard for every detector, and why? Which detector misses most DoS rows, and why does a flood fool it?
5. **(D2)** How does the shrinkage λ change the Mahalanobis result? Why might the λ chosen by Ledoit-Wolf not be the best for detection?
6. **(E1)** What score does a point get when it is isolated after 2 cuts, and after 20? Why do the trees use small subsamples?
7. **(F1)** For which unseen category does the forest fail? What does that say about deploying only a classifier?
8. **(F2)** Propose a combination of a classifier and a one-class detector for a real network, and say who labels the alarms.

Detector	AUPRC	dos	probe	r2l	u2r
Mahalanobis (your $\lambda = 0.01$)					
Mahalanobis (Ledoit-Wolf)					
best of the zoo					
random forest without r2l					

Optional challenges

- **Robust covariance:** fit `MinCovDet` on 5,000 training rows of the numeric features only. Does it beat Ledoit-Wolf? Add 1% of attacks from the pool to the training rows and compare again.
- **Contamination:** repeat part D with 1% and 5% of the pool mixed into the training rows. Which detector degrades least (notebook 02, part C)?
- **GMM size:** sweep the number of components of the Gaussian mixture (2 to 40). Is there a best value, and how would you choose it without labelled attacks?

Reference solution

The solved notebook, with the answers to the questions and the numbers of a full run, is `solution/guide_03_01.ipynb`.